

Target: Specialist Programmer (₹21 LPA / ₹16 LPA)

Don't prepare for the ₹6.25 LPA role. Prepare for the highest level. If you can solve L3-level questions, you'll also be competitive for the other roles.

Phase 1: Master DSA (Highest Priority)

These topics cover the majority of coding interview questions.

Week 1

Arrays

- Two Sum
- Best Time to Buy and Sell Stock
- Remove Duplicates
- Rotate Array
- Kadane's Algorithm
- Merge Intervals

Practice:

- Easy: 10
- Medium: 15

Strings

- Reverse String
- Valid Palindrome
- Longest Common Prefix
- Longest Substring Without Repeating Characters
- Group Anagrams
- Minimum Window Substring

HashMap & HashSet

- Frequency counting
- First Unique Character
- Top K Frequent Elements
- Contains Duplicate
- Happy Number

Sliding Window

Must know:

- Maximum Sum Subarray
- Longest Repeating Character Replacement
- Minimum Window Substring
- Fruits Into Baskets

Week 2

Binary Search

Must solve:

- Binary Search
- Search Insert Position
- First and Last Position
- Peak Element
- Search in Rotated Array

Linked List

- Reverse Linked List
- Merge Two Lists
- Detect Cycle
- Remove Nth Node
- Middle Node

Stack

- Valid Parentheses
- Min Stack
- Next Greater Element
- Daily Temperatures

Queue

- Implement Queue
- Circular Queue
- Sliding Window Maximum

Week 3

Trees

Most Important

- Tree Traversals
- Height

- Diameter
- Balanced Tree
- Lowest Common Ancestor
- Level Order Traversal
- BST Search
- BST Insert

Heap / Priority Queue

- K Largest Elements
- Merge K Sorted Lists
- Top K Frequent
- K Closest Points

Recursion

- Factorial
- Fibonacci
- Subsets
- Permutations
- Combination Sum

Week 4

Graphs

- DFS
- BFS
- Number of Islands
- Flood Fill
- Detect Cycle
- Topological Sort

Dynamic Programming

Don't skip.

Learn

- Climbing Stairs
- House Robber
- Coin Change
- Longest Increasing Subsequence
- Longest Common Subsequence
- 0/1 Knapsack

Java Revision

Know these thoroughly:

Collections

ArrayList

LinkedList

HashMap

HashSet

TreeMap

PriorityQueue

Deque

Stack

Important APIs

`Arrays.sort()`

`Collections.sort()`

`Collections.reverse()`

`Collections.binarySearch()`

StringBuilder

Comparator

SQL Revision

Practice

JOIN

LEFT JOIN

RIGHT JOIN

GROUP BY

HAVING

ORDER BY

LIMIT

COUNT

SUM

AVG

MIN

MAX

WINDOW FUNCTIONS

SUBQUERY

DBMS

Study

Normalization

ACID

Transactions

Locks

Deadlock

Indexing

Primary Key

Foreign Key

Views

Triggers

Operating Systems

Must Know

Process

Thread

CPU Scheduling

Deadlock

Semaphore

Mutex

Paging

Virtual Memory

Computer Networks

Important

OSI

TCP/IP

HTTP

HTTPS

DNS

DHCP

Cookies

Sessions

REST

OOP

Be able to explain:

Encapsulation

Inheritance

Polymorphism

Abstraction

Interface

Abstract Class

Method Overloading

Method Overriding

SOLID Principles

JavaScript / TypeScript

Since you've used them in projects, be ready for questions on:

Promises

Async/Await

Closures

Event Loop

Hoisting

this keyword

Arrow Functions

Modules

TypeScript Interfaces

Generics

Projects

Be ready to explain every project on your resume using this structure:

- Problem statement
- Why you chose the technology
- System architecture
- Database design
- API flow
- Challenges faced
- Optimizations made
- Future improvements

Expect follow-up questions like:

- "Why did you choose SQL over NoSQL?"
- "How would your application scale to 1 million users?"
- "How did you secure your APIs?"

HR Interview

Prepare answers for:

- Tell me about yourself.
- Why Infosys?
- Why should we hire you?
- Biggest strength?
- Biggest weakness?
- Leadership experience?
- Conflict resolution?
- Are you willing to relocate?
- Five-year career plan?

Coding Practice Goal

Try to solve:

- **150–200 DSA problems** before the assessment:
 - 60 Easy
 - 100 Medium
 - 40 Hard (or as many as you can comfortably attempt)

Focus on quality over quantity—understand the patterns behind each solution.

Daily Schedule (4–5 Hours)

Time	Activity
1.5 hours	DSA learning + implementation
1.5 hours	Timed coding practice
45 minutes	CS fundamentals (OS/DBMS/CN/OOP)
30 minutes	SQL practice
30 minutes	Java revision

My Plan to Help You

I can mentor you from now until the Infosys assessment. Here's the roadmap I'd suggest:

Phase 1: DSA Foundation (15 days)

- Daily lessons
- Curated problems

- Solutions and pattern explanations

Phase 2: Advanced Coding (10 days)

- Graphs
- Dynamic Programming
- Hard interview problems

Phase 3: Mock Assessments (5 days)

- 3-hour coding tests that simulate the Infosys assessment
- Performance analysis
- Personalized feedback

Phase 4: Interview Preparation

- Technical mock interviews
- HR mock interviews
- Resume review and project discussions

I recommend starting with Day 1: Arrays, where we'll cover:

1. Time complexity and Big-O basics.
2. Array patterns used in interviews.
3. Five essential problems with increasing difficulty.
4. A short coding assignment.
5. A timed mini-test to build speed.

By the end of this program, you'll have practiced the core interview patterns and be much better prepared for the Specialist Programmer assessment.

Since you're targeting **Infosys Specialist Programmer (₹21 LPA / ₹16 LPA)**, you don't need hundreds of resources. The biggest mistake students make is jumping between platforms. Instead, use a small set of high-quality resources consistently. Completing one structured DSA sheet (such as NeetCode 150 or Striver's A2Z) is generally more effective than switching between multiple lists.



1. DSA (Highest Priority)

1. Striver's TakeUForward ★★★★★ (Must Do)

Website:

[TakeUForward \(Striver\)](#)

YouTube:

takeUforward YouTube Channel

Recommended:

- A2Z DSA Course

- SDE Sheet
- Graph Series
- Dynamic Programming Series

2. NeetCode

Website:

[NeetCode](#)

YouTube:

NeetCode YouTube Channel

Start with:

- NeetCode 150
- Roadmap
- Pattern-based learning

3. LeetCode

Website:

[LeetCode](#)

Practice:

- Top Interview 150
- LeetCode 75
- Daily Challenge

4. GeeksforGeeks

Website:

[GeeksforGeeks](#)

Best for:

- Theory
- DSA implementation
- Interview experiences



Java

Kunal Kushwaha (Java + DSA)

YouTube:

Kunal Kushwaha YouTube Channel

Excellent for:

- Java
- Collections Framework
- OOP
- DSA

Bro Code

Bro Code YouTube Channel

Great for quick Java revisions.



SQLBolt

[SQLBolt](#)

Interactive lessons from beginner to advanced.

HackerRank SQL

[HackerRank SQL Practice](#)

LeetCode Database

[LeetCode Database Problems](#)



Gate Smashers

Gate Smashers YouTube Channel

Jenny's Lectures

Jenny's Lectures CS/IT



Gate Smashers

Same channel—excellent OS playlist.

Neso Academy

Neso Academy YouTube Channel



Computer Networks

Gate Smashers

Neso Academy

Both provide placement-focused explanations.



OOP

Kunal Kushwaha

Bro Code

Both explain OOP concepts clearly with Java examples.



Web Development (Useful for Interviews)

MDN Web Docs

[MDN Web Docs](https://developer.mozilla.org/en-US/docs/Web)

JavaScript.info

[JavaScript.info](https://javascript.info)

TypeScript Handbook

[TypeScript Handbook](#)



System Design (Good to Know)

ByteByteGo

[ByteByteGo](#)



Interview Experiences

GeeksforGeeks

[GeeksforGeeks Interview Experiences](#)

LeetCode Discuss

[LeetCode Discuss](#)



Resume

Overleaf (ATS-friendly templates)

[Overleaf Resume Templates](#)



Coding Contests

- [Codeforces](#)
- [CodeChef](#)
- [AtCoder](#)



YouTube Channels Worth Following

- takeUforward (Striver)
- NeetCode
- Kunal Kushwaha
- Gate Smashers
- Neso Academy

- Bro Code

My Recommended Path (in order)

1. **Striver A2Z DSA** (primary roadmap)
2. **NeetCode 150** (pattern recognition)
3. **LeetCode** (daily practice)
4. **SQLBolt + LeetCode Database**
5. **Gate Smashers** (DBMS, OS, CN)
6. **MDN and TypeScript Handbook** for web fundamentals

If your goal is specifically **Infosys Specialist Programmer**, this combination is more than sufficient and avoids resource overload while covering the topics most commonly tested.

also give me some github repo that have all the infosys related questions

Yes! I've shortlisted the **best GitHub repositories** that are actually useful for preparing for **Infosys Specialist Programmer (SP)** and **Digital Specialist Engineer (DSE)**. These contain previous coding questions, company-wise interview problems, and solutions.

★ 1. Infosys SP Coding Questions (Highly Recommended)

GitHub:

[karthikreddy-7/Infosys-SP-Coding-Questions](https://github.com/karthikreddy-7/Infosys-SP-Coding-Questions)

Why it's useful

- Dedicated to Infosys Specialist Programmer preparation
- Questions categorized by topic:
 - Arrays
 - Strings
 - Greedy
 - Trees
 - Graphs
 - Dynamic Programming
- Java solutions
- Similar difficulty to recent SP assessments

★ 2. PacketPrep Coding Questions

GitHub:

[packetprep/coding-questions](https://github.com/packetprep/coding-questions)

Contains:

- Infosys

- TCS
- Capgemini
- Cognizant
- Wipro
- Accenture

Includes more than **200 coding questions** from placement drives with explanations. Great for mastering the common logic-building problems asked in service-company assessments.

★ 3. Programming Aptitude Interview Prep

GitHub:

[SAKET-SK/Programming-Aptitude-Interview-Prep](https://github.com/SAKET-SK/Programming-Aptitude-Interview-Prep)

Contains:

- Company-wise folders
- Infosys questions
- Java solutions
- C++
- Python
- Interview preparation notes

Excellent if you want one repository covering multiple companies while still including Infosys-style questions.

★ 4. InfyTQ Coding Questions

GitHub Topic Page:

[GitHub Infosys/InfyTQ repositories](https://github.com/Infosys/InfyTQ-repositories)

Look for repositories such as:

- InfyTq-Coding-Questions
- InfyTq-2022-Preparation

These focus on previous InfyTQ and HackWithInfy coding problems, many of which overlap with Specialist Programmer preparation.

★ 5. NeetCode (Must Complete)

GitHub:

[NeetCode GitHub](https://github.com/NeetCode)

Although it's not Infosys-specific, many recent interview experiences report that the coding round emphasizes patterns like arrays, strings, trees, graphs, and DP—exactly the areas NeetCode covers.

★ 6. Striver's A2Z DSA Sheet

GitHub:

Striver SDE Sheet GitHub

If you finish this sheet, you'll cover nearly all the core DSA patterns commonly discussed in Infosys SP interviews.

★ 7. Blind 75

GitHub:

[Blind 75 LeetCode GitHub](#)

A concise set of interview problems that strengthens your fundamentals.

★ 8. Company-wise LeetCode Repository

GitHub:

[LeetCode Company-wise Questions](#)

Useful if you later prepare for companies beyond Infosys, such as Amazon, Microsoft, Google, or Adobe.



Official Preparation Pages

Besides GitHub, keep these bookmarked:

- [GeeksforGeeks Infosys SDE Sheet](#) — Curated Infosys interview questions and coding topics.
- [TakeUForward Infosys Interview Experience](#) — Real interview experiences and preparation guidance.

🎯 If I were preparing for Infosys SP 21 LPA today

I would follow this order:

1.  Striver A2Z DSA Sheet
2.  NeetCode 150
3.  Infosys-SP-Coding-Questions (GitHub)

4.  PacketPrep Coding Repository
5.  GeeksforGeeks Infosys SDE Sheet
6.  Daily LeetCode (2–3 problems)

That combination covers the topics most frequently reported in recent Infosys SP/DSE coding rounds—especially arrays, strings, linked lists, trees, graphs, greedy algorithms, and dynamic programming.

 **I can also create a single GitHub repository for you containing:**

-  300+ Infosys previous coding questions
-  Java solutions
-  PDFs and notes
-  Aptitude questions
-  DBMS, OS, CN, and OOP notes
-  Interview experiences
-  A 30-day preparation tracker

It would be organized in one place so you don't have to search across multiple websites.

also give me some other links that will help full for my prep

Since you're aiming for **Infosys Specialist Programmer (₹21 LPA)**, I've put together a **curated list of the best free resources**. These are the same resources used by many students preparing for product-based companies as well.



1. Complete Placement Roadmaps



Developer Roadmap

<https://roadmap.sh>

Best for:

- Backend
- Frontend
- Java
- System Design
- SQL
- Git
- DevOps

HackWithInfy Preparation Hub

<https://hwi-prep-guide.vercel.app/>

This is one of the most complete free guides specifically for **HackWithInfy** and **Infosys Specialist Programmer**, including:

- Real coding questions (2024–2026)
- Java solutions
- DSA roadmap
- Theory questions
- Interview preparation
- 6-week study plan.

DSA Sheets

Striver A2Z Sheet ★★★★★

<https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2>

Striver SDE Sheet

<https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/>

NeetCode 150

<https://neetcode.io/roadmap>

Sean Prashad LeetCode Patterns

<https://seanprashad.com/leetcode-patterns/>

Excellent for learning by problem pattern.

CSES Problem Set

<https://cses.fi/problemset/>

One of the best free collections for algorithm practice.

Coding Platforms

LeetCode

<https://leetcode.com/problemset/>

Codeforces

<https://codeforces.com>

CodeChef

<https://www.codechef.com>

AtCoder

<https://atcoder.jp>

HackerRank

<https://www.hackerrank.com>

Great for:

- SQL
- Java
- Problem Solving



CS Fundamentals

GeeksforGeeks

<https://www.geeksforgeeks.org>

Best for:

- DBMS
- OS
- CN
- OOP
- SQL
- Aptitude

TutorialsPoint

<https://www.tutorialspoint.com>

Great for quick revision.

Java Documentation

<https://docs.oracle.com/en/java/>



SQL Practice

SQLBolt

<https://sqlbolt.com>

SQLZoo

<https://sqlzoo.net>

LeetCode SQL

<https://leetcode.com/problemset/database/>



Web Development

MDN

<https://developer.mozilla.org>

JavaScript.info

<https://javascript.info>

TypeScript Handbook

<https://www.typescriptlang.org/docs/>



YouTube Playlists

Striver

<https://www.youtube.com/@takeUforward>

NeetCode

<https://www.youtube.com/@NeetCode>

Kunal Kushwaha

<https://www.youtube.com/@KunalKushwaha>

Gate Smashers

<https://www.youtube.com/@GateSmashers>

Perfect for:

- DBMS
- OS
- CN

Abdul Bari

https://www.youtube.com/@abdul_bari

Best for:

- Algorithms
- Graphs

Aditya Verma

<https://www.youtube.com/@TheAdityaVerma>

Must-watch for:

- Dynamic Programming

William Fiset

<https://www.youtube.com/@WilliamFiset-videos>

Best Graph tutorials available.



Cheat Sheets

Big-O Cheat Sheet

<https://www.bigocheatsheet.com>

Refactoring Guru

<https://refactoring.guru>

Excellent for:

- Design Patterns
- OOP



System Design

ByteByteGo

<https://bytebytego.com>

System Design Primer (GitHub)

<https://github.com/donnemartin/system-design-primer>

One of the most-starred system design repositories.



Aptitude

IndiaBIX

<https://www.indiabix.com>

Perfect for:

- Quantitative Aptitude
- Logical Reasoning
- Verbal Ability

FreshersNow

<https://www.freshersnow.com>

Contains company-specific aptitude questions.



Interview Experiences

GeeksforGeeks

<https://www.geeksforgeeks.org/company-interview-corner/>

LeetCode Discuss

<https://leetcode.com/discuss/>

Reddit Infosys Community

<https://www.reddit.com/r/infosys/>

Recent interview experiences suggest that after the coding round, interviewers often focus on OOP, DBMS, OS, CN, SQL, and an in-depth discussion of your projects, so don't limit your preparation to DSA alone.



GitHub Repositories

Awesome Interview Questions

<https://github.com/DopplerHQ/awesome-interview-questions>

Tech Interview Handbook

<https://github.com/yangshun/tech-interview-handbook>

Probably the best interview preparation repository on GitHub.

Coding Interview University

<https://github.com/jwasham/coding-interview-university>

A complete Computer Science roadmap.

The Algorithms

<https://github.com/TheAlgorithms/Java>

Algorithms implemented in Java.

Java Design Patterns

<https://github.com/luwatar/java-design-patterns>

A must-read for Java interviews.

Awesome Java

<https://github.com/akullpp/awesome-java>

Collection of Java resources.



My Recommended Daily Stack

If I had **30 days** to prepare for the Infosys SP role, I'd use only these:

Purpose	Resource
DSA Learning	Striver A2Z
DSA Practice	LeetCode
Pattern Recognition	NeetCode 150
SQL	SQLBolt + LeetCode Database
DBMS/OS/CN	Gate Smashers + GeeksforGeeks
Java	Kunal Kushwaha
System Design	System Design Primer
Aptitude	IndiaBIX
Interview Experiences	Reddit + GeeksforGeeks
Infosys-Specific	HackWithInfy Preparation Hub



Bonus: "Ultimate Placement Preparation" GitHub Repositories

If you're looking for repositories that collect almost everything in one place, these are excellent:

- **Tech Interview Handbook** – <https://github.com/yangshun/tech-interview-handbook>
- **Coding Interview University** – <https://github.com/jwasham/coding-interview-university>
- **Complete Placement Preparation** – <https://github.com/anushka23g/Complete-Placement-Preparation>
- **DSA KIT** – <https://github.com/avinash201199/DSA-KIT>

These repositories bring together roadmaps, DSA sheets, CS fundamentals, interview questions, and placement resources in a structured way, making them a good complement to the Infosys-specific materials.